Sampling Distributions

Def: If $X_1, \ldots, X_n$ are independent and each has the same (marginal) distribution with CDF F (or pdf/pmf f), we say $X_1, \ldots, X_n$ are IID and write $X_1, \ldots, X_n \sim F$ (of f).
We call $X_1, \ldots, X_n$ a random sample of size n from F
(CB Def 5.1.1)

Def: Let $T: \mathbb{R}^n \to \mathbb{R}^d$. The random vector/variable $Y = T(X_1, \ldots, X_n)$ is called a statistic. The probability distribution of Y is called the sampling distribution (CB Def 5.2.1)

Common statistics: $\bar{X}$, S^2, median, IQR, S, proportions p, odds, t, ...

Example: If $X_1, \ldots, X_n \sim$ Bernoulli(p), then the sampling distribution of $Y = \sum_{j=1}^{n} X_j$ is Binomial(n, p) (STAT2003 - section 3.4)

↳ parametrised by mean & variance

Question: If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, what is the sampling distribution of $\bar{X}$?

$$\mathbb{E}(\bar{X}) = \mathbb{E}\left(n^{-1} \sum_{j=1}^{n} X_j\right) = n^{-1} \sum_{j=1}^{n} \mathbb{E}(X_j) = n^{-1} \cdot n\mu = \mu.$$

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(n^{-1} \sum_{j=1}^{n} X_j\right) = n^{-2}\, \mathrm{Var}\left(\sum_{j=1}^{n} X_j\right) = n^{-2} \sum_{j=1}^{n} \mathrm{Var}(X_j)$$

$$= n^{-2} \cdot n\sigma^2 = \sigma^2/n.$$

A sum of independent normal random variables has a normal distribution. $\bar{X} \sim N(\mu, \sigma^2/n)$ (STAT2003 - claim 41)

Questions: If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, what is the sampling distribution of $(\bar{X}, S^2)$?

Multivariate Normal Distribution

An m-dimensional random vector W is said to have a multivariate normal distribution (MVN) if for every $t \in \mathbb{R}^m$ $t'W = t_1 W_1 + \cdots + t_m W_m$ has a (univariate) normal distribution.
The MVN distribution is parametrised by mean vector μ and covariance Σ

$$\mu_j = \mathbb{E}(W_j) \quad \text{and} \quad \Sigma_{ij} = \text{cov}(W_i, W_j)$$

(STAT2003 - section 7.6)

- If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, then $(X_1, \ldots, X_n)' \sim N(\mu \mathbf{1}, \sigma^2 I)$
  (follows Claim 41 - STAT2003)

- Suppose $W = \begin{pmatrix} U \\ V \end{pmatrix}$. If every component of U is uncorrelated with every component of V, the U and V are independent. (expanding on Claim 40 - STAT2003)

Independence of $\bar{X}$ and S^2. $(X_1, \ldots, X_n \sim N(\mu, \sigma^2))$

Recall $\bar{X} = n^{-1} \sum_{j=1}^{n} X_j$ and $S^2 = (n-1)^{-1} \sum_{j=1}^{n} (X_j - \bar{X})^2$.

1) The random vector $W = (X_1, \ldots, X_n)'$ has a MVN distribution

2) The random vector $(\bar{X}, X_1 - \bar{X}, \ldots, X_n - \bar{X})$ has a MVN distribution since we can write this a BW where B is an $(n+1) \times n$ matrix (constants). So any $t \in \mathbb{R}^m$ $t'(BW) = (t'B)W$ since $(t'B)' \in \mathbb{R}^n$ and W has a MVN distribution

3) What is the covariance between $\bar{X}$ and $X_j - \bar{X}$.
If U, V, W are random variables and $a, b \in \mathbb{R}$, the
$\text{cov}(aU + bV, W) = a\,\text{cov}(U, W) + b\,\text{cov}(V, W)$
$\text{cov}(U, V) = \text{cov}(V, U)$

$$\begin{aligned}\operatorname{cov}(X_j-\bar{X},\bar{X}) &= \operatorname{cov}(X_j,\bar{X}) - \operatorname{cov}(\bar{X},\bar{X}) \\ &= \operatorname{cov}(X_j, n^{-1}X_1+\cdots+n^{-1}X_n) - \operatorname{var}(\bar{X}) \\ &= n^{-1}\operatorname{cov}(X_j,X_1)+\cdots+n^{-1}\operatorname{cov}(X_j,X_n) - \operatorname{var}(\bar{X}) \\ &= n^{-1}\operatorname{cov}(X_j,X_j) - \operatorname{var}(\bar{X}) \\ &= \sigma^2/n - \sigma^2/n = 0\end{aligned}$$

$$\operatorname{cov}(X_i,X_j) = \begin{cases} 0, & \text{if } i\neq j \\ \sigma^2, & \text{if } i=j \end{cases}$$

So $\bar{X}$ is uncorrelated with every component of $(X_1-\bar{X},\ldots,X_n-\bar{X})$

(ii) Since uncorrelated implies independence within a MVN vector, we have $\bar{X}$ is independent of $(X_1-\bar{X},\cdots,X_n-\bar{X})$. As S^2 is a function of $(X_1-\bar{X},\ldots,X_n-\bar{X})$, we see S^2 is independent of $\bar{X}$.

Distribution of S^2

(chi-squared distribution – STAT2003 pg 225)

Recall (?) If $Z_1,\ldots,Z_k \sim N(0,1)$, then $Q = Z_1^2+\cdots+Z_k^2$ has a χ^2_k distribution.

$$\mathbb{E}Q = k, \quad \operatorname{Var}(Q) = 2k \quad \text{and} \quad M_Q(t) = (1-2t)^{-k/2}$$
$$t < 1/2.$$

If U and V are independent rvs, then

$$M_{U+V}(t) = M_U(t)\cdot M_V(t)$$

Moment generating functions are unique (STAT2003 – claim 48) so if we can identify the MGF of S^2, then we know its distribution.

Suppose $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$. Then $\frac{X_j - \mu}{\sigma} \sim N(0,1)$

and then $\sum_{j=1}^{n} \left(\frac{X_j - \mu}{\sigma}\right)^2 = \frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \mu)^2 \sim \chi^2_n$

We can write

$$\frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \mu)^2 = \frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \bar{X} + \bar{X} - \mu)^2$$

$$= \frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \bar{X})^2 + \frac{2}{\sigma^2} \sum_{j=1}^{n} (X_j - \bar{X})(\bar{X} - \mu) + \frac{1}{\sigma^2} \sum_{j=1}^{n} (\bar{X} - \mu)^2$$

$(\bar{X} - \mu)$ is constant, $= 0$

$$= \frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \bar{X})^2 + \frac{n(\bar{X} - \mu)^2}{\sigma^2}$$

Independent

As $\frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \sim N(0,1)$ so $\frac{n(\bar{X} - \mu)^2}{\sigma^2} \sim \chi^2_1$

Using MGFs,

$$M_{\frac{1}{\sigma^2}\sum_{j=1}^{n}(X_j - \mu)^2}(t) = M_{\frac{1}{\sigma^2}\sum_{j=1}^{n}(X_j - \bar{X})^2}(t) \cdot M_{\frac{n(\bar{X}-\mu)^2}{\sigma^2}}(t)$$

$$(1-2t)^{-n/2} = M_{\frac{1}{\sigma^2}\sum_{j=1}^{n}(X_j - \bar{X})^2}(t) \cdot (1-2t)^{-1/2}$$

$$M_{\frac{1}{\sigma^2}\sum_{j=1}^{n}(X_j - \bar{X})^2}(t) = (1-2t)^{-(n-1)/2}$$

which is the MGF of a χ^2_{n-1} distribution.

$$\frac{1}{\sigma^2} \sum_{j=1}^{n} (X_j - \bar{X})^2 = \frac{(n-1)}{\sigma^2} S^2 \sim \chi^2_{n-1}.$$

Point Estimators and their evaluation.

Given $X_1, \ldots, X_n \sim F$, how do we infer F, or some feature $\psi(F)$.

Def: A point estimator is any function $T(X_1, \ldots, X_n)$ of the random sample. (CB Def 7.1.1)

$\underbrace{T(\overbrace{X_1, \ldots, X_n}^{\text{random sample}})}_{\text{Estimator}}$ $\qquad$ $\underbrace{T(\overbrace{x_1, \ldots, x_n}^{\text{observed values}})}_{\text{Estimate.}}$

Def: A statistical model $\mathcal{F}$ is a set of distributions.

Def: Parametric model $\mathcal{F} = \{ f(\cdot;\theta) : \theta \in \Theta \subseteq \mathbb{R}^d \}$ where for each $\theta \in \Theta$, $f(\cdot;\theta)$ is a pdf / pmf. Here θ is called the parameter and Θ is the parameter space.

* **Important:** We assume that distinct values of θ correspond to distinct distributions.

Example (Bernoulli) $\quad \mathcal{F} = \{ f(x;p) = p^x (1-p)^{1-x}, \; p \in [0,1] \}$
$x \in \{0,1\}$

Example (Normal) $\quad \mathcal{F} = \{ f(x;\mu,\sigma^2) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$
$\mu \in \mathbb{R}, \sigma^2 > 0 \}$

So $\theta = (\mu, \sigma^2)$ and $\Theta = \mathbb{R} \times (0, \infty)$

Method of moments

We estimate $\theta \in \Theta$ by equating the first d sample moments with the moments of the distribution.

$k = 1, \ldots, d \qquad m_k = \frac{1}{n} \sum_{j=1}^{n} X_j^k \qquad \alpha_k(\theta) = \mathbb{E}_\theta (X^k)$

$= \int x^k f(x;\theta)\, dx$ (in continuous case)

provided they exist

Example (Bernoulli) If $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, then

$$\alpha_1(p) = \mathbb{E}_p(X) = p \qquad \text{so} \qquad \hat{p} = \frac{1}{n}\sum_{j=1}^{n} X_j$$

Example (Normal) If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, then

$$\alpha_1(\theta) = \mathbb{E}_\theta(X) = \mu \qquad m_1 = \frac{1}{n}\sum_{j=1}^{n} X_j = \bar{X}$$

$$\alpha_2(\theta) = \mathbb{E}_\theta(X^2) = \sigma^2 + \mu^2 \qquad m_2 = \frac{1}{n}\sum_{j=1}^{n} X_j^2$$

Recall $\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$

$\sigma^2 = \mathbb{E}(X^2) - \mu^2$

$$\hat{\mu} = \bar{X}$$

$$\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum_{j=1}^{n} X_j^2$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{j=1}^{n} X_j^2 - \bar{X}^2$$

$$= \frac{1}{n}\sum_{j=1}^{n} (X_j - \bar{X})^2$$

(not quite the sample variance)

Example (Geometric distribution)

$$f(x;p) = p(1-p)^{x-1}, \quad x = 1, 2, 3, \dots \quad p \in [0,1]$$

It can be shown that

$$\alpha_1(p) = \mathbb{E}_p(X) = 1/p \qquad \text{(STAT2003 - claim 13)}$$

$$1/p = \frac{1}{n}\sum_{j=1}^{n} X_j = \bar{X} \quad \Rightarrow \quad \hat{p} = 1/\bar{X}$$

Example (Zero inflated Poisson)

$$f(x;\theta) = \begin{cases} \pi + (1-\pi)e^{-\lambda}, & x = 0 \\ (1-\pi)\dfrac{e^{-\lambda}\lambda^x}{x!}, & x = 1, 2, \dots \end{cases}$$

(Poisson)

$\theta = (\pi, \lambda) \in [0,1] \times (0, \infty)$

$$\alpha_1(\theta) = \mathbb{E}_\theta(X) = (1-\pi)\lambda \qquad m_1 = (1-\pi)\lambda$$

$$\alpha_2(\theta) = \mathbb{E}_\theta(X^2) = (1-\pi)(\lambda + \lambda^2) \qquad m_2 = (1-\pi)(\lambda + \lambda^2)$$

$$\frac{m_2}{m_1} = \frac{(1-\pi)(\lambda+\lambda^2)}{(1-\pi)\lambda} = \lambda+1 \quad \Rightarrow \hat{\lambda} = \frac{m_2}{m_1} - 1$$

$$m_1 = (1-\pi)\lambda \quad \Rightarrow$$

$$(1-\pi) = \frac{m_1}{\lambda} \quad \Rightarrow \hat{\pi} = 1 - \frac{m_1}{\hat{\lambda}} = 1 - \frac{m_1^2}{m_2 - m_1}$$

Do these estimators always lie in the parameter space?

Is $\hat{\lambda} > 0$? We would need $m_2 > m_1$

This always holds for rvs taking values in $\{0, 1, 2, \ldots\}$

Is $\hat{\pi} \in (0,1)$ $\quad \hat{\pi} = \dfrac{m_2 - m_1 + m_1^2}{m_2 - m_1}$ (estimate variance)

Note that $m_2 - m_1^2$ is an estimator of variance

$\hat{\pi} \in (0,1)$ if $m_2 - m_1^2 > m_1$